Multi-Key Recall in Converted Hybrid Language Models: A Training Deficit, Not a Capacity Limit

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Abstract

Several recipes convert a pretrained transformer into a hybrid whose linear-attention or state-space layers have a fixed-size memory. We ask what this does to long-context associative recall and whether the loss can be repaired. With a synthetic suite that plants 16 facts in documents of 1K to 32K tokens, we compare Jet-Nemotron-2B and an 8B Mamba-in-the-Llama model with the transformers they came from. Both hybrids pass a single-needle test up to 16K (0.86 and 1.00 at 32K), but at 32K recall one of 16 planted facts at 0.31 and 0.25, against 0.90 and 0.99 for their sources; RULER's multi-key tasks show the same gap. We then train a low-rank adapter on under 1% of the weights on synthetic many-fact documents for 42 to 47 minutes on one GPU. Recall at 32K returns to the source model's level (0.99 and 1.00, means over three seeds); the 8B adapter was trained only up to 16K. The deficit is thus one of training, not of capacity of the converted model as deployed, on the suite at both scales and on RULER's numeric tasks at 8B; at 2B a gap to the identically trained source on RULER remains, and whether the fixed-size states alone could hold the facts is untested. The repair transfers to a held-out fact template at both scales. Copying the source's query and key projections does not recover recall. Its limits follow its training data and scale: at 2B, multi-hop questions were learned only when in the mix; UUID values, absent from the data, are repaired at 8B within the training length and at 2B not at all, though the identically trained source improves modestly; multi-answer questions, also absent, are partly repaired at 8B. Collateral effects are small at 2B (+0.2 to +0.4% bits per byte on long text, benchmarks within one standard error); at 8B the 2B dose over-fits (+15%) and a lighter one gives full recall at +5.5 to +7.5%. We release the harness, training code and raw outputs.

1 Introduction

Full attention stores a key and value for every token in the context, so the memory needed to serve a request grows linearly with its length. Linear-attention and state-space layers instead maintain a state of fixed size, and hybrid architectures that keep a small number of full-attention layers alongside such layers now serve long contexts at a fraction of the memory cost [1, 2]. Because training a hybrid from scratch is expensive, several recipes convert an existing pretrained transformer into a hybrid, or into a fully recurrent model, by distillation [3–7]. These papers evaluate the converted models mainly on short-context language-modelling benchmarks; long-context evaluation, where present, is a single-needle retrieval test or an aggregate score.

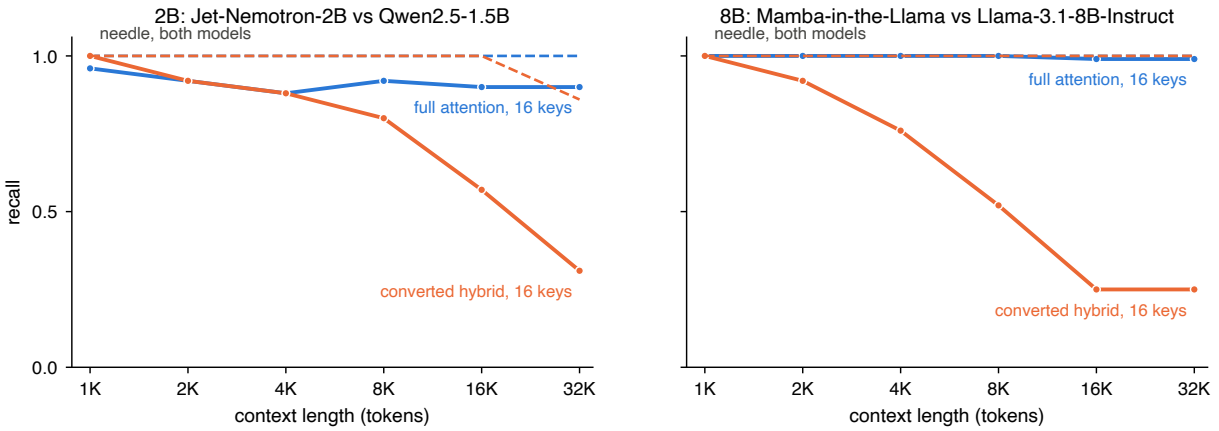


Figure 1: Recall as a function of context length: 25 cases per point at 1K to 8K, 100 at 16K and 32K. Dashed lines: single needle. Solid lines: one of 16 random keys. Each converted hybrid is paired with the transformer it was converted from.

Two lines of prior work suggest that this evaluation may miss something. Models with a fixed-size recurrent state are known to lose associative recall and copying ability as the number of stored facts and the distance to them grow [8–13]. And a single needle [14] is the easiest case for such a state, which is the reason the RULER benchmark includes variants with several simultaneous needles [15].

This paper asks three questions about converted hybrids specifically.

1. Relative to the exact transformer it was converted from, does a converted hybrid lose multi-key recall at long context while still passing the single-needle test?
2. If so, is the deficit a capacity limit of the converted model, or a training deficit? By capacity we mean that of the converted model as deployed, its retained attention layers and its fixed-size states together. A small adapter trained after conversion distinguishes the two: if training restores recall, the capacity was present. We do not ask whether the recurrent states alone could hold the facts.
3. What does such a repair cost in general language-modelling ability, and what are its boundaries?

Our contributions are as follows. (i) A paired measurement of two converted hybrids against their source transformers at 1K to 32K tokens, which shows a large multi-key recall deficit alongside single-needle recall of 1.00 up to 16K at both scales (0.86 and 1.00 at 32K) (Figure 1, Section 4.1). (ii) A post-hoc low-rank adapter that restores multi-key recall at 32K to the level of the source model, replicated over three seeds at each scale, with a full-attention control trained on the same data at 2B, transfer to a held-out fact template at both scales and to RULER's numeric multi-key tasks, nearly fully at 8B and partly at 2B, generalisation past the training length at 8B, and a training-free baseline that copies the source's query and key projections and fails (Sections 4.2 and 4.3). (iii) An analysis of what the repair reaches beyond its training data: at 2B the hybrid learned multi-hop questions only once the mix contained them; at 8B the repair transfers when one thing is new, UUID values within the training length or numeric values beyond it, barely when both are (UUID at 32K, 0.24 against 0.96), and partly for multi-answer questions beyond the training length; at 2B UUID recall does not improve while the identically trained source's improves modestly

(Section 5.1); together with a measurement of the collateral cost and a recipe that controls it at 8B (Section 5.2).

2 Related work

Converting transformers into hybrids. Mamba-in-the-Llama distils Llama-3 and Zephyr models into hybrids in which half to all of the attention layers are replaced by Mamba or Mamba-2 layers, using the attention projections to initialise the new layers [3, 16]. LoLCATs linearises Llama and Mistral models by training linear attentions to mimic softmax attention, then correcting the approximation error with LoRA [4]. SUPRA replaces the softmax attention of Llama and Mistral models with linear attention and uptrains them [5]. RADLADS converts Qwen2.5 models into RWKV-style recurrent decoders with a three-stage distillation [6]. Jet-Nemotron selects which attention layers to keep with a search over a pretrained model, replaces the rest with a linear-attention block, and then trains the full model [7]. Lizard converts Llama-3-8B while keeping half of its softmax attention layers and reports RULER parity with its teacher up to 32K [17]. The LoLCATs authors report that their 8B conversion loses passkey retrieval at 8K and recover it by using passkey data as the conversion data itself [4, App. B.5.2]; LoLA later repairs single-needle retrieval in LoLCATs models at inference time with a sparse cache, while their multi-key RULER scores remain low [18].

Recall in fixed-state models. Zoology and Based measure associative recall in efficient language models and characterise the trade-off between recall and state size [8, 9]. An empirical study of Mamba-based models at 8B scale finds that pure Mamba models trail transformers on copying and phonebook lookup, and that adding a few attention layers recovers much of the gap [11]. Stuffed Mamba characterises state collapse beyond the training length and the state capacity for passkey retrieval [13]. Transformers outperform state-space models at copying from the context, in theory and in practice [10]. Just Read Twice narrows the recall gap of recurrent models at inference by repeating the context, without training [12]; our repair is trained once and leaves the prompt and the inference cost unchanged. In transformers, retrieval from the context is carried by a small set of attention heads [19], and the same gather-and-aggregate mechanism, concentrated in few heads, appears in state-space models and hybrids [20]; this bears on our finding that, at 8B and on our suite, an adapter on the retained attention layers' query and key projections recovers most of the recall (Section 5.2). MiniMax report returning from a hybrid to full attention in their M2 model after observing deficits in multi-hop reasoning that small-scale benchmarks had not revealed [21].

Long-context evaluation. RULER extends the single needle-in-a-haystack test with multi-key, multi-value and multi-query variants, variable tracking and aggregation tasks, and has become the standard synthetic evaluation of effective context length [15]. An unpaired RULER aggregate has been reported for Jet-Nemotron-2B, falling to 46.6 at 32K [22], and a third party reports Jet-Nemotron-2B between 57.2 and 100 on RULER's three single-needle variants at 32K [23]. Neither work compares the converted model with the model it was converted from, and neither reports multi-key scores separately.

Recall training and forgetting. Continued training of from-scratch Mamba-transformer hybrids on recall-oriented data improves recall at a few thousand tokens with a small cost in commonsense accuracy; training on needle-in-a-haystack data specifically is reported to harm commonsense

benchmarks [24]. In a different setting, fine-tuning retrieval components for LLM agents on synthetic data with LoRA causes forgetting on real and out-of-distribution data [25]. Concurrently with this work, Zhou et al. [23] show that chain-of-thought fine-tuning degrades long-range recall in hybrids and attribute the damage to drift in the query and key projections, which they repair by restoring those projections from the pre-fine-tuning checkpoint. Our setting differs in that we repair a deficit introduced by conversion rather than by later fine-tuning, and we do so with a trained adapter; our observation that, at 8B and on our suite, an adapter restricted to the query and key projections recovers most of the recall at the lowest bits-per-byte cost among the recipes that keep multi-hop recall (Section 5.2) is consistent with their explanation. Section 4.3 applies their training-free restore with the source transformer as the reference and finds that it does not carry over to the conversion setting.

3 Experimental setup

3.1 Models

We study two converted hybrids, each paired with the transformer it was converted from. Jet-Nemotron-2B [7] has 28 blocks, of which 2 use full attention and 2 use sliding-window attention; the remaining 24 use the linear-attention JetBlock. It was initialised from Qwen2.5-1.5B with the MLPs frozen during the search stage; the final stage then trained the whole model for 350B tokens. Qwen2.5-1.5B is therefore the initialisation checkpoint rather than a weight-identical base, a caveat we return to in Section 6. Mamba-in-the-Llama 8B is the released `Llama3.1-Mamba2-8B-dpo` checkpoint [3], in which 16 of the 32 layers of Llama-3.1-8B-Instruct [26] are replaced by Mamba-2 layers [16]. Its comparison model is Llama-3.1-8B-Instruct. All models are run in bfloat16 with PyTorch's scaled-dot-product attention on NVIDIA A100-40GB GPUs, except the 8B gentle, replay and query-and-key-only runs of Section 5.2, which ran on A40 GPUs.

3.2 Synthetic recall suite

Each test document consists of filler text with planted sentences. Filler sentences are drawn deterministically from a combinatorial bank of 32^3 sentences about office logistics and contain no digits. A planted sentence has the form *The access code for locker KX-402 is 481930*; values are six-digit codes, chosen so that no two values in a document share their first three digits. Because the filler contains no digits, a six-digit number in a completion can only have been copied from a planted sentence or invented. Documents are sized under a single reference tokenizer (Qwen2.5) to targets of 1K, 2K, 4K, 8K, 16K and 32K tokens, landing within 1.3% of the target; every model receives byte-identical prompts.

We use five tasks. *Needle* plants a single fact, as in the standard needle-in-a-haystack test, and serves as the control. *16 random keys* plants sixteen facts with unrelated keys and asks for one of them. *16 similar keys* plants sixteen facts whose keys share a letter prefix and carry sixteen consecutive numbers (for example KX-401 to KX-416). *Updated* plants eight facts, each later overwritten by a sentence that changes its value, and asks for the current value; it is run but not reported in the tables, because the full-attention sources often return the overwritten value (at 32K Qwen2.5-1.5B returns the overwritten value in 45 of 100 cases), so it measures instruction following rather than recall; at 32K the untrained models score 0.16 to 0.74 on it and every adapter 0.96 or above. *Multi-hop* plants eight two-step chains (a locker is assigned to a person; the person has a

code) and asks for the code of the person assigned to a given locker. A sixth task, *parcels*, plants sixteen facts in a different template (*Parcel 48213-K has the pickup code 731902*) and is used only for evaluation, never for training; it shares the filler bank and the instruction with the training documents, so only the fact sentence and the key format are held out.

Prompts are completion-style, with no chat template: an instruction, the document, the question and the token *Answer:*. Decoding is greedy with 32 new tokens. A case is scored as recalled if the planted value appears among the six-digit numbers on the first non-empty line of the completion. Scoring on the first number alone changes one reported rate, Llama-3.1-8B-Instruct on multi-hop at 32K (0.46 instead of 0.49), which lists several codes on one line in 62 of 100 cases; 4 of the 372 evaluated cells differ in all, the other three unreported. Guessing among the planted codes gives 0.06 on the sixteen-key tasks and 0.13 on multi-hop. Rates are rounded half up. Unless stated otherwise, results use 25 cases per task and length in the initial comparison and 100 cases per task and length (two independently generated suites of 50) in the repair experiments. The standard error of a rate is at most 0.10, 0.07 and 0.05 at 25, 50 and 100 cases; where several training seeds exist we report their mean or their range.

3.3 RULER

To relate our suite to a standard benchmark we also run six needle variants from RULER [15], generated with the authors' released code and the same reference tokenizer: single needle in a noise haystack, three multi-key variants (three distractor keys in an essay haystack; a haystack composed of needles; UUID keys and values), multi-value and multi-query, with 50 cases per task at 16K and 32K. Scoring follows RULER's rule: a case is correct if every expected string appears anywhere in the completion. Completions are limited to 64 new tokens. The models rarely stop earlier on the multi-value and multi-query tasks, so nearly every completion of those tasks reaches the limit at both scales, mostly while repeating a value, listing wrong ones or, for the untrained 8B hybrid, drifting into other numbers from the haystack. A miss that reaches the limit with fewer than four distinct values and no repeated value, and that ends inside a number or within 40 characters of its last value, may be an answer cut short. Such misses make up at most 0.40 of a cell for the untrained 2B models (the 16K multi-query cell of Jet-Nemotron-2B; its 16K multi-value cell has none), 0.36 for the repaired 2B models (at 32K; at most 0.14 for multi-value at 16K), 0.32 for the untrained 8B hybrid and 0.04 for the repaired 8B hybrid. The multi-value and multi-query rates are therefore lower bounds: loose at 2B and for the untrained 8B hybrid, tight for the repaired 8B hybrid and for the 8B source. At 2B we draw conclusions only from multi-value at 16K. RULER prompts were sized with the Qwen2.5 tokenizer; under Llama's tokenizer, which encodes three digits per token, the 8B models see about 16K and 32K tokens on the single-needle, multi-value and multi-query prompts, 13K and 26K on the second multi-key variant and 12K and 25K on the UUID variant, so the 8B UUID results at the nominal 16K lie within the adapter's 8K to 16K training lengths and those at 32K beyond them.

3.4 Repair procedure

The repair data are synthetic documents of the same construction as the suite, restricted to the locker template. Each document plants facts of one of four kinds, cycled evenly: 16 to 48 random keys, 16 to 48 near-duplicate keys, 8 to 24 keys each overwritten once, or 8 to 24 two-hop chains; it ends with eight questions and their answers. Our first experiments used only the first three kinds; the fourth was added after the analysis in Section 5.1. The training loss is the cross-entropy of the

answer tokens only. Because a 16K-token document with a 128K-token vocabulary produces 8 GiB of logits, the output head is applied only at the answer positions, which keeps training within a single 40 GB GPU.

We train a LoRA adapter [27] of rank 16 and $\alpha = 32$ on every linear projection outside the MLPs, that is, the attention projections and the projections of the linear-attention or Mamba-2 layers; the embeddings, output head and MLPs are frozen. At 2B this wraps 232 layers and 0.75% of the parameters; at 8B, 96 layers and 0.15%. Optimisation uses AdamW with four documents per step, a constant learning rate without warm-up, gradient clipping at 1.0, no dropout, no weight decay except 0.01 in the 50% replay run at 5×10^{-5} (150 + 150 documents, Section 5.2), and gradient checkpointing; the adapter factors are kept in float32 on a bfloat16 model. We report two training doses. The *heavy* recipe uses 600 documents (150 steps) at learning rate 2×10^{-4}; the *gentle* recipe uses 300 documents at 5×10^{-5}. The 2B model is trained on documents of 8K, 16K and 32K tokens; the 8B model only on 8K and 16K, so that its results at 32K measure generalisation beyond the training length. The heavy 2B run processes 11.5M tokens in 42 minutes on one A100 (measured on a run of the same dose with the original three-kind mix; the reported seeds add two-hop documents of the same lengths); the gentle 8B runs process 3.7M tokens in 43–47 minutes on one A40 (three seeds, from the job logs). After training, the adapter is merged into the weights, so inference cost is unchanged; merged and unmerged evaluation agree on all but 12 of 3,600 cases (twelve adapters, 300 cases each at 32K).

3.5 Controls and collateral measurements

To separate repair from mere familiarity with the prompt format, the same recipe is applied to the full-attention source model at 2B; a gain that appears in both models cannot be attributed to repair alone; a gain only in the hybrid is repair. To measure collateral effects we report ARC-Easy and HellaSwag accuracy on 500 validation items each (standard error 0.022), and bits per byte on eight windows of 16K tokens from a long natural text (*Moby-Dick*), before and after repair. The recipes of Section 5.2 were compared on the same suites we report; there is no separate selection set. Generation was found to be deterministic: 150 of 150 completions on the parcel-template suite were byte-identical between two independent runs on the same GPU type.

4 Results

4.1 Converted hybrids lose multi-key recall while passing the needle test

Figure 1 plots recall against context length for both pairs, and Table 1 gives the values at 32K together with RULER's multi-key tasks. In both, the values at 16K and 32K come from the evaluation with 100 cases per cell and those at 1K to 8K from the initial sweep with 25. The 25-case sweep itself gave different values at these lengths: for the 2B pair at 32K, 0.72 for the source and 0.16 for the hybrid, and for the 8B hybrid 0.48 at 16K and 0.12 at 32K against 0.25 and 0.25 with 100 cases, differences of about two standard errors that we attribute to the small sample. All four models score 1.00 on the single-needle test at every length up to 16K; at 32K the sources and the 8B hybrid remain at 1.00 and Jet-Nemotron-2B scores 0.86 (1.00 in the 25-case sweep). On sixteen planted facts the sources remain near their short-context level, whereas the hybrids degrade with length. At 2B the gap opens at 16K (0.57 against 0.90) and reaches 0.59 at 32K; at 8B it opens at 4K and is near total at 32K, where the hybrid recalls one of sixteen facts at 0.25 against 0.99 for

Table 1: Recall at 32K before repair. Suite columns use 100 cases per cell; RULER columns use 50. MK1–MK3: RULER multi-key variants 1–3; MV: multi-value, a lower bound under the 64-token generation limit, loose at 2B (Section 3.3). Parcels is the held-out fact template.

	needle	16 random	16 similar	multi-hop	parcels	MK1	MK2	MK3	MV
Qwen2.5-1.5B, source	1.00	0.90	0.75	0.14	0.94	0.94	0.72	0.40	0.46
Jet-Nemotron-2B, hybrid	0.86	0.31	0.22	0.11	0.45	0.56	0.26	0.16	0.02
Llama-3.1-8B-Instruct, source	1.00	0.99	1.00	0.49	1.00	1.00	1.00	0.96	0.98
Mamba-in-the-Llama 8B, hybrid	1.00	0.25	0.09	0.06	0.17	0.40	0.10	0.00	0.00

Llama-3.1-8B-Instruct. The misses are well formed: in every case we inspected, the model returned another planted code in a grammatical answer.

RULER's multi-key tasks detect the same failure (Table 1), so our suite adds no detection power over RULER; its role here is to provide the paired measurement and the training signal. Two further observations run against our expectations. At 2B, near-duplicate keys were no harder than random keys below 32K; at 8B they were harder from 8K on (0.08 against 0.52 at 8K in the 25-case sweep), so key geometry matters at 8B but is not the dominant stressor at 2B. And the 8B conversion, which keeps half of its attention layers, loses recall earlier and more completely than the 2B conversion, which keeps two full-attention and two sliding-window layers out of 28; the number of retained attention layers does not by itself predict the deficit.

4.2 A small adapter restores recall

Figure 2 and Table 2 show recall at 32K before and after repair. At 2B, three seeds of the heavy recipe raise recall on sixteen random keys from 0.31 to 1.00, 0.98 and 1.00, and on sixteen similar keys from 0.22 to 1.00, 0.98 and 1.00. The full-attention control trained on the same data reaches 1.00 and 0.97, so the repaired hybrid matches the trained source. The control also bears on whether the gains are repair or familiarity with the prompt format. The untrained source already scores 0.90 and 0.75 under byte-identical prompts, and the hybrids score 1.00 on the single needle in the same format up to 16K (Jet-Nemotron-2B 0.86 at 32K), so the format is not the obstacle; the control's training moves it by 0.10 and 0.22 to the ceiling, while the hybrid moves by 0.68 and 0.77. Because the source starts near the ceiling, the control cannot bound how much of the hybrid's gain is familiarity. On multi-hop, where the control starts low, both models gain (control 0.14 to 0.99, hybrid 0.11 to 0.86), so the training teaches two-hop retrieval that the untrained source also lacked, and the hybrid's gain cannot be split into repair and general learning. At 8B, three seeds of the gentle recipe raise recall on random keys from 0.25 to 1.00, 1.00 and 0.99, and on similar keys from 0.09 to 0.99, 1.00 and 0.99, matching the untrained Llama-3.1-8B-Instruct. These 8B adapters were trained only on documents of 8K and 16K tokens, so the repair generalises beyond the training length on the numeric tasks; Section 5.1 shows where it does not.

The repair also transfers. On the held-out parcel template, which differs from the training data in fact wording and key format but shares its filler text, recall at 32K rises from 0.45 to 0.93–0.98 at 2B and from 0.17 to 0.98–1.00 at 8B. On RULER, whose prompts use a different instruction, an essay or needle-only haystack and seven-digit values, the 8B hybrid's multi-key scores at 32K rise from 0.40 and 0.10 to 0.96–1.00 and 0.86–0.90 on the first two variants, at or approaching the source model's 1.00; at 2B they rise from 0.56 and 0.26 to 0.81 and 0.49 (means over seeds), against 0.94 and 0.72 for the source and 0.96 and 0.97 for the trained control, so at 2B the transfer to RULER is

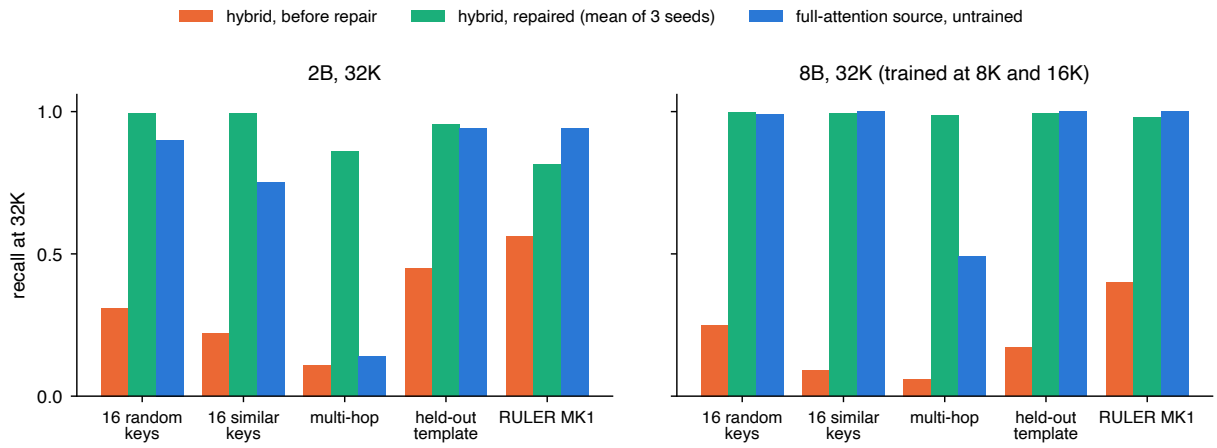


Figure 2: Recall at 32K before and after repair, next to the untrained full-attention source, whose 2B values are in Table 1; Table 2 gives the trained control at 2B. Repaired values are means over three training seeds of the heavy recipe at 2B and the gentle recipe at 8B. The 8B adapter was trained only on documents up to 16K tokens.

Table 2: Recall at 32K after repair. Suite columns use 100 cases per cell and RULER columns 50; repaired values are means over three seeds. At 2B the control is the source model trained with the same recipe (mean over two seeds); at 8B the source was not retrained. MV is a lower bound under the 64-token generation limit, loose at 2B (Section 3.3). Parcels is the held-out fact template.

		16 random	16 similar	multi-hop	parcels	MK1	MK2	MV
2B	hybrid, before	0.31	0.22	0.11	0.45	0.56	0.26	0.02
	hybrid, repaired (heavy)	0.99	0.99	0.86	0.96	0.81	0.49	0.12
	source, trained (control)	1.00	0.97	0.99	1.00	0.96	0.97	0.37
8B	hybrid, before	0.25	0.09	0.06	0.17	0.40	0.10	0.00
	hybrid, repaired (gentle)	1.00	0.99	0.99	0.99	0.98	0.89	0.73
	source, untrained	0.99	1.00	0.49	1.00	1.00	1.00	0.98

partial and the repaired hybrid stays below the identically trained control.

We conclude that the deficit is a training deficit rather than a capacity limit of the converted model: with its retained attention layers and its fixed-size states, it can recall sixteen facts over 32K tokens once trained to do so, and four to twelve million tokens of targeted training supply the retrieval the conversion did not. The inference holds for the sixteen-key and held-out tasks at both scales, for multi-hop at 8B, and for RULER's numeric multi-key tasks at 8B. At 2B the repaired hybrid stays below the identically trained control on multi-hop (0.86 against 0.99) and on RULER's multi-key tasks at 16K and 32K (at 32K 0.81, 0.49 and 0.19 against 0.96, 0.97 and 0.57, Appendix B), and whether that residual gap is a training or a capacity limit is not resolved here. The result does not locate the restored retrieval. The adapter also trains the retained softmax-attention layers, whose memory grows with context, and at 8B an adapter on their query and key projections alone recovers most of the recall on our suite (Section 5.2), so those layers may carry it; whether the recurrent states alone could, we do not test.

4.3 A training-free baseline: copying the source's query and key projections

Attention Amnesia [23] repairs recall lost during chain-of-thought fine-tuning without training: it copies the query and key projections of the retained attention layers back from the checkpoint before fine-tuning. A converted hybrid has no such checkpoint, but the source transformer is the natural candidate, since the hybrid was initialised from it. We therefore copied the source's query and key projections into every retained softmax-attention layer of each hybrid (four layers at 2B, two full-attention and two sliding-window; sixteen at 8B) and ran the same evaluation. This follows their Algorithm 1, which copies the query and key weights of each listed softmax-attention layer, with the source transformer in place of the pre-fine-tuning checkpoint; we also copy the biases, which Qwen2.5-1.5B has, and neither model has query or key normalisation.

The copy is inapplicable at 2B and harmful at 8B. At 2B it removes retrieval altogether: recall is 0.00 on every task at 16K and 32K (100 cases per cell); on 93% of the 1,200 cases, the single needle included, the model answers with a sentence saying the answer is not in the text, and on the rest with a repetition loop; no completion contains a planted code, and four contain a placeholder string of digits such as 1234567890. The reason is visible in the weights. Jet-Nemotron-2B's softmax-attention projections have cosine similarity below 0.01 with Qwen2.5-1.5B's at the same layer and at any other layer, and below 0.09 under any pairing of heads, while its MLPs (0.71) and embeddings (0.88) still resemble the source. Whether the recipe re-initialised the attention layers or trained them far from their start, they carry nothing of Qwen's weights, so no reference checkpoint exists for them. The method's own Jet-Nemotron-2B experiment, with the released checkpoint as the reference, also recovers only part of the fine-tuning loss (their single needle in an essay haystack, RULER's NIAH-Single-2, at 32K: 57.8 to 64.6, against 95.6 before fine-tuning). At 8B the retained attention layers do stay close to Llama-3.1-8B-Instruct's (cosine 0.98–0.99 on the query and key projections), yet copying them lowers recall at 32K on sixteen random keys from 0.25 to 0.04 and on the single needle from 1.00 to 0.72 (56–60 cases per cell at 32K: 50 from the first suite and 6–10 from the second, whose 32K run was stopped early; 100 cases at 16K give 0.07 and 0.67), where the adapter gives 1.00 on both (needle not tabulated; see the released outputs). One reading is that the sixteen attention layers have adapted to the Mamba-2 layers between them, so that the source's projections no longer fit; we did not test it. The contrast with Section 5.2, where an adapter restricted to the same query and key projections but trained on the recall data recovers most of the deficit, shows that, at 8B and on our suite, those projections can carry most of the repair, but that their values must be learned in the converted model.

5 Analysis

5.1 What the repair reaches beyond its training data

The adapter restores mainly the kind of retrieval its training data contains, with transfer beyond it that depends on scale. The positive direction, that adding a question type to the mix teaches it, was tested for multi-hop questions at 2B. For UUID keys and values and for multi-answer questions, which the training data lacks, we observe transfer at 8B; at 2B the UUID task does not move, and the multi-answer results are inconclusive under the generation limit except for multi-value at 16K (Section 3.3). Appendix B tabulates every RULER variant.

Multi-hop questions. Our first training mix contained only one-hop questions. The resulting 2B adapter answered two-hop questions with the person's name, that is, the one-hop value adjacent to the locker key, in 124 of the 150 cases of the 25-case sweep (all six lengths pooled); multi-hop recall

over those cases fell from 0.17 to 0.02, and at 32K from 0.20 to 0.08, while the control's rose from 0.16 to 0.39 pooled over lengths (0.16 to 0.28 at 32K). Adding two-hop chains to the mix raised multi-hop recall at 32K to 0.86 at 2B and 0.99 at 8B; no 8B adapter was trained without them.

UUID keys and values. The training data contains only numeric values. At 8B, RULER's third multi-key variant, with UUID keys and values, is repaired within the training lengths (its nominal 16K prompts are about 12K tokens under the model's tokenizer, Section 3.3), 0.10 to 0.80–0.98 against 0.98 for Llama-3.1-8B-Instruct, but barely beyond them (about 25K tokens): 0.00 to 0.08–0.32 against 0.96. The numeric multi-key tasks, whose value type the training data contains, generalise to 32K (Section 4.2); the UUID task, whose value type it lacks, does not. At 2B the picture is different. The untrained hybrid has no UUID deficit at 16K (0.64 against 0.40 for the source) and a deficit only at 32K (0.16 against 0.40). Training does not move it (0.50–0.62 at 16K, 0.18–0.22 at 32K), while the control, trained on the same number-only data, improves at 32K from 0.40 to 0.64 and 0.50 (mean 0.57, about two standard errors) and at 16K from 0.40 to 0.62 and 0.56. The absence of UUIDs from the training data therefore does not by itself explain the 2B result: the same data improve UUID retrieval in the full-attention model, modestly, and not in the hybrid.

Multi-answer questions. Every training question has exactly one answer. At 8B, RULER's multi-value and multi-query tasks are near the source at 16K (0.78–0.88 and 0.90–0.94 against 0.92 and 1.00) and partly repaired at 32K (0.68–0.80 and 0.60–0.78 against 0.98 and 1.00). At 2B, multi-value recovers at 16K to the trained control's level (0.24 to 0.48–0.70, against 0.74 for the untrained source and 0.54–0.60 for the trained control, which loses on both multi-answer tasks: 0.74 to 0.54–0.60 and 0.64 to 0.52–0.56). The untrained multi-query rate at 16K (0.08) is a loose lower bound, up to 0.40 of that cell being answers cut short, so its rise to 0.36–0.62 cannot be read as recovery. At 32K the measured rates stay low (0.06–0.22 and 0.02–0.08) and up to 0.36 of each cell may be cut short, so we draw no conclusion there.

So at 8B the repair transfers when one thing is new, UUID values within the training length or numeric values beyond it, barely when both are, and partly for multi-answer questions beyond the training length. At 2B, UUID recall does not improve while the identically trained source's improves modestly, and the multi-answer results are inconclusive except for multi-value at 16K. A practical repair should therefore contain the question types, value types and lengths of the intended workload. We expect, but did not test, that adding UUID and multi-answer questions to the mix would close the remaining gaps at 8B; at 2B the UUID gap at 32K is part of the unresolved residual of Section 4.2.

5.2 Collateral cost and the training dose

At 2B the collateral effects are small. ARC-Easy and HellaSwag accuracy change by less than the standard error of a single estimate (0.022) for all three seeds, and bits per byte on long natural text rise from 1.082 to 1.084–1.087 (+0.2% to +0.4%); the trained control pays a similar cost (+0.4% to +0.6%). A paired test on the same items was not run, because per-item results were not stored; Appendix A lists every value.

At 8B the same recipe over-fits: the training loss reaches zero, bits per byte rise by 15% (from 1.008 to 1.156) and HellaSwag falls by 3.4 points. Figure 3 and Table 3 compare the recipes we tried. Reducing the dose to 300 documents at a quarter of the learning rate (the gentle recipe) gives full recall with bits per byte at +5.5% to +7.5% and benchmarks within noise, consistently over three seeds. Restricting the adapter to the query and key projections, at the heavy dose, reduces the cost to +1.7% with no benchmark change; this variant recalls well on our suite but transfers

less to RULER than the full adapter (MK1 0.72 against 0.98). We recommend the gentle recipe as the default and report its cost rather than claiming none.

Replay removes the cost but takes retrieval with it. Interleaving windows of natural text (*War and Peace*, disjoint from the evaluation text) with ordinary language-modelling loss, a standard remedy for forgetting, lowers bits per byte below the unrepaired model, but in all three replay variants multi-hop recall falls from 0.99 to 0.30–0.36, similar-key recall to 0.77–0.98 and RULER's first multi-key variant to 0.64–0.82, while random-key recall stays at 0.99 or above (Table 3, Figure 3). The three variants span 150 to 600 synthetic documents and two learning rates and give similar multi-hop scores, so we have no dose-response curve. One hypothesis is that the task needing the most retrieval signal is the first to go when part of the signal is replaced; we did not test it.

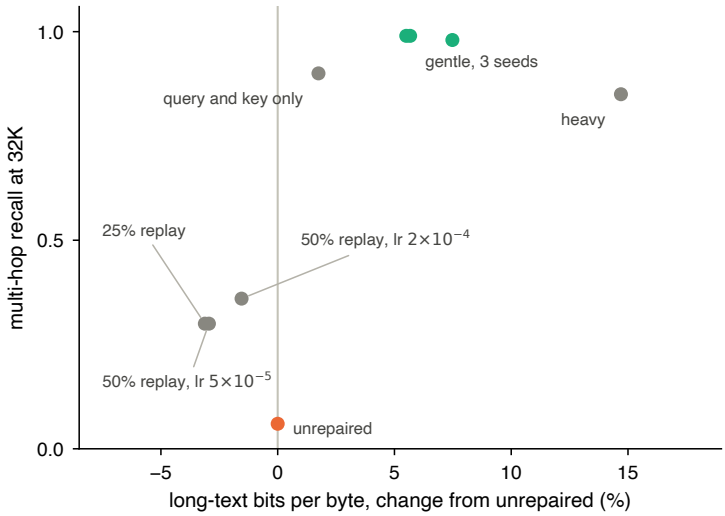


Figure 3: Training recipes at 8B: change in bits per byte on long natural text against multi-hop recall at 32K. Each grey or green point is one training run, the gentle recipe for three seeds; the orange point is the unrepaired model.

Table 3: Training recipes at 8B. Documents: synthetic repair documents plus natural-text windows seen in training; the learning rate is constant. Recall at 32K uses 100 cases; RULER MK1 and MK3 at 32K use 50; bits per byte is measured on 16K windows of long natural text; HellaSwag accuracy is on 500 items (standard error 0.022). In the gentle row, recall and RULER are means over three seeds and bits per byte and HellaSwag are ranges. Percentage changes in the text are computed from unrounded values.

recipe	documents	lr	16 random	16 similar	multi-hop	MK1	MK3	bits/byte	HellaSwag
unrepaired	–	–	0.25	0.09	0.06	0.40	0.00	1.008	0.600
heavy	600 + 0	2×10^{-4}	0.98	0.91	0.85	1.00	0.30	1.156	0.566
gentle (3 seeds)	300 + 0	5×10^{-5}	1.00	0.99	0.99	0.98	0.24	1.063–1.083	0.584–0.594
25% replay	600 + 200	5×10^{-5}	1.00	0.98	0.30	0.82	0.12	0.976	0.578
50% replay, lr 5×10^{-5}	150 + 150	5×10^{-5}	0.99	0.77	0.30	0.64	0.06	0.978	0.582
50% replay, lr 2×10^{-4}	300 + 300	2×10^{-4}	0.99	0.86	0.36	0.70	0.12	0.992	0.574
query and key only	600 + 0	2×10^{-4}	1.00	0.89	0.90	0.72	0.02	1.026	0.606

6 Limitations

We evaluate two conversion families at two scales; other conversions may behave differently, and Lizard's reported RULER parity [17] shows that the deficit is not universal. Neither source is a weight-identical base: Jet-Nemotron-2B's attention layers carry nothing of Qwen2.5-1.5B's and its MLPs and embeddings only partly resemble Qwen's (cosine 0.71 and 0.88, Section 4.3), and the 8B hybrid's retained layers, though close to Llama-3.1-8B-Instruct's (0.98 to 0.99), were trained further, so part of each gap may come from that training rather than from the attention replacement itself; the repair results do not depend on this attribution. The 8B recipe comparison uses one training seed per recipe except for the gentle recipe, and the trained full-attention control exists only at 2B. Our collateral measurements are ARC-Easy, HellaSwag and bits per byte on one book; we do not evaluate a natural long-context task such as document question answering. At 8B, UUID recall beyond the training length stays far below the source under every recipe (Table 3), and multi-answer recall below it; multi-answer rates are lower bounds under the 64-token generation limit, and at 2B the bound is loose enough that the multi-answer results are inconclusive except for multi-value at 16K (Section 3.3). At 2B the repaired hybrid also stays below the identically trained control on multi-hop and on RULER's multi-key tasks at 16K and 32K, and we do not resolve whether that residual gap is a training or a capacity limit. At 8B an adapter on the retained attention layers' query and key projections alone recovers most of the recall on our suite (Table 3), so those layers may carry the restored retrieval; whether the recurrent layers alone can, we do not test. Finally, our suite is synthetic; whether the repaired recall carries over to retrieval from real documents is not tested here.

7 Conclusion

Converted hybrid language models can pass the single-needle test up to 16K (0.86 and 1.00 at 32K) while recalling one of sixteen planted facts at 32K at a quarter to a third of the rate of the transformers they were converted from. A low-rank adapter trained for under an hour on synthetic many-fact documents restores recall to the source model's level at both 2B and 8B, at 8B generalises past its training length, and transfers to a held-out fact template at both scales and, nearly fully at 8B, to RULER's numeric multi-key tasks. The deficit is therefore a training deficit rather than a capacity limit of the converted model as deployed, on the suite at both scales and on RULER's numeric tasks at 8B; at 2B the repaired hybrid stays below the identically trained control on multi-hop and on RULER, a residual we do not resolve. The restored retrieval may run through the retained attention layers; we did not locate it. The repair is limited by its training data and by scale: at 2B, multi-hop questions were learned only once the mix contained them; UUID keys and values, absent from the data, were repaired at 8B within the training length and at 2B not at all, while the identically trained source improved modestly; multi-answer questions, also absent, were partly repaired at 8B. Its cost is small at 2B and controllable at 8B. Copying the source transformer's query and key projections, the training-free route, does not work in this setting: at 2B the hybrid's attention layers carry nothing of the source's weights, and at 8B the copy lowers recall further. These results suggest that conversion pipelines should include a recall-repair stage, and that any such stage should be built from the question and value types of the workload it is meant to serve.

Reproducibility. The evaluation harness, the training code, the Slurm job scripts, every raw completion and the scripts that produce each table and figure are available at `https://github.`

`com/gottn/recall-guard`. The harness evaluates any Hugging Face model with a single command and reports a recall curve by length.

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A Collateral measurements

Table 4: ARC-Easy and HellaSwag accuracy (500 items each, standard error 0.022) and bits per byte on eight 16K windows of *Moby-Dick* (mean and standard error over windows) for every model and adapter in the paper. 2B adapters use the heavy recipe; the 8B rows name the recipe of Table 3.

model	ARC-Easy	HellaSwag	bits/byte
Qwen2.5-1.5B, untrained	0.746	0.520	1.064 ± 0.018
Qwen2.5-1.5B, trained, seed 1	0.736	0.520	1.070 ± 0.018
Qwen2.5-1.5B, trained, seed 2	0.732	0.518	1.068 ± 0.018
Jet-Nemotron-2B, untrained	0.764	0.508	1.082 ± 0.017
Jet-Nemotron-2B, repaired, seed 1	0.776	0.492	1.086 ± 0.017
Jet-Nemotron-2B, repaired, seed 2	0.770	0.510	1.087 ± 0.018
Jet-Nemotron-2B, repaired, seed 3	0.770	0.498	1.084 ± 0.017
Llama-3.1-8B-Instruct, untrained	0.808	0.598	0.800 ± 0.035
Mamba-in-the-Llama 8B, untrained	0.848	0.600	1.008 ± 0.021
Mamba-in-the-Llama 8B, heavy	0.836	0.566	1.156 ± 0.020
Mamba-in-the-Llama 8B, gentle, seed 1	0.842	0.586	1.065 ± 0.021
Mamba-in-the-Llama 8B, gentle, seed 2	0.854	0.594	1.063 ± 0.021
Mamba-in-the-Llama 8B, gentle, seed 3	0.832	0.584	1.083 ± 0.020
Mamba-in-the-Llama 8B, 25% replay	0.834	0.578	0.976 ± 0.020
Mamba-in-the-Llama 8B, 50% replay, lr 5×10^{-5}	0.826	0.582	0.978 ± 0.020
Mamba-in-the-Llama 8B, 50% replay, lr 2×10^{-4}	0.822	0.574	0.992 ± 0.021
Mamba-in-the-Llama 8B, query and key projections only	0.848	0.606	1.026 ± 0.021

B RULER, all variants

Table 5: RULER needle variants at 16K and 32K, 50 cases per cell, means over seeds where several exist. S1: single needle in a noise haystack; MK1–MK3: multi-key variants 1–3; MV: multi-value; MQ: multi-query. Multi-value and multi-query rates are lower bounds under the 64-token generation limit, loose at 2B and for the untrained 8B hybrid (Section 3.3).

model	length	S1	MK1	MK2	MK3	MV	MQ
Qwen2.5-1.5B, untrained	16K	1.00	0.98	0.86	0.40	0.74	0.64
	32K	1.00	0.94	0.72	0.40	0.46	0.48
Qwen2.5-1.5B, trained (2 seeds)	16K	1.00	1.00	0.98	0.59	0.57	0.54
	32K	1.00	0.96	0.97	0.57	0.37	0.50
Jet-Nemotron-2B, untrained	16K	1.00	0.72	0.60	0.64	0.24	0.08
	32K	1.00	0.56	0.26	0.16	0.02	0.00
Jet-Nemotron-2B, repaired (3 seeds)	16K	1.00	0.85	0.86	0.57	0.62	0.49
	32K	1.00	0.81	0.49	0.19	0.12	0.05
Llama-3.1-8B-Instruct, untrained	16K	1.00	1.00	1.00	0.98	0.92	1.00
	32K	1.00	1.00	1.00	0.96	0.98	1.00
Mamba-in-the-Llama 8B, untrained	16K	1.00	0.72	0.40	0.10	0.32	0.14
	32K	1.00	0.40	0.10	0.00	0.00	0.00
Mamba-in-the-Llama 8B, gentle (3 seeds)	16K	1.00	1.00	0.96	0.91	0.83	0.91
	32K	1.00	0.98	0.89	0.24	0.73	0.71